

Abhrajeet Mukherjee

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PROFESSIONAL SUMMARY

Software Engineer - AI Backend experienced in building **LLM-powered backend systems**, **agentic workflows**, scalable APIs, async orchestration, RAG pipelines, and enterprise AI automation platforms.

EDUCATION

BITS Pilani: *Bachelor of Engineering*, Computer Science, 2025 CGPA: 7.17
Ryan International School, Mumbai: Class XII 95%
Delhi Public School, Varanasi: Class X 94.2%
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, OOP

EXPERIENCE

Software Developer - Full Stack AI Engineer June 2025 – Present
73 Strings Bengaluru

- Built AI backend systems for PE valuation workflows across KKR, Blackstone, Apollo, enabling scalable portfolio automation
- Redesigned portfolio aggregation using acyclic tree-based metrics rollup engines for hierarchical fund and fund-of-funds analytics
- Contributed for core Sidebar Agent to Trigger platform workflows and Tool based Insights via LLM based orchestration
- Built Sheriff validation agent for business-rule governance, workflow reliability, and automated financial data quality checks
- Scaled async orchestration systems from 20 to 150 jobs with ~50% memory utilization across distributed enterprise services

Software Engineer Intern January 2025 – June 2025
73 Strings Bengaluru

- Built backend APIs and enterprise workflows supporting document processing, validation, and financial data operations
- Created a Node.js migration pipeline enabling seamless data portability between PROD environments across 4 enterprise products
- Developed Angular-based financial workflow modules for NAV computation and real-time portfolio monitoring
- Reduced API response time by 90% using optimized compression, payload reduction, and backend processing improvements

Full Stack Engineer Intern Jul 2024 – Dec 2024
Unity Growth Fund Remote

- Built and maintained a private equity investment platform focused on portfolio monitoring and investor performance analytics
- Developed scalable modules using React, Redux, Node.js, and Flutter for real-time financial workflows
- Designed LoopBack 4 APIs powering 15+ investment workflow features including valuation and fund distribution tracking
- Implemented S3-based document workflows with tagging and retrieval pipelines for operational efficiency

LEADERSHIP

Chief Coordinator and Events Head | *May 2022 – Jun 2023* Quark Controls

- Scaled Quark, the Annual Techno-Management fest, increasing revenue by 58% through strategic growth initiatives
- Led a 250+ member cross-functional team managing operations, stakeholders, and execution workflows
- Managed a 1+ crore budget using data-driven operational planning and allocation strategies

PROJECTS

Voice Sentiment Analysis Pipeline | *Applied ML Research*

- Built speech intelligence pipelines using MFCCs, formants, F0 tracking, and RMS energy profiling workflows
- Combined acoustic embeddings with ASR-driven semantic sentiment and contextual inference classification workflows
- Implemented prosodic analysis for speaking rate variability, tonal stress, and pause segmentation workflows
- Designed adaptive vocal baseline models for emotional deviation and conversational confidence scoring systems

Agentic Sales Call Analysis Platform | *Conversational AI Analytics*

- Built AI intelligence pipelines extracting objections, buying intent, and conversational behavior analysis signals
- Designed hybrid speech and LLM reasoning systems generating contextual coaching intelligence workflows

Marketstate - Plot Visualization and Broker Management Software | *Product & Full Stack Engineering*

- Built immersive 3D visualization systems enabling interactive plot exploration and guided walkthrough workflows
- Deployed production-ready real estate platforms improving customer engagement and remote buyer discovery

TECHNICAL SKILLS

Gen AI Systems: AI Agents, RAG Pipelines, LLM Orchestration, Prompt Engineering, Tool Calling, MCP, Vector Search, AI Workflows, Evals, Multimodal AI
ML Frameworks: PyTorch, Scikit-learn, Transformers, TensorFlow, Librosa, Pandas, NumPy
Backend Frameworks: Spring Boot, Node.js, FastAPI
Frontend Frameworks: React, Next.js, Angular
Data & Infrastructure: AWS, Azure, PostgreSQL, Redis, RabbitMQ, Elasticsearch, Docker
Programming Languages: C++, Python, Java